Appendix B.3

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Overview

Descriptive full-text title

Assay to test impairment of migration of human neural crest cells (cMINC; UKN2) – V2.0

Abstract

This in vitro test method uses human neural crest cells (NCCs) generated from induced pluripotent stem cells (iPSC). It assesses disturbances of NCC migration during fetal development. The number of migrated NCCs, as well as cell viability are measured simultaneously using high content imaging. Thereby the processes of cell migration and cell death are measured in cells exposed to potential toxicants for 24 h. The data of this method are meant to predict developmental disorders and malformations e.g., neural tube defects or craniofacial malformations caused by compound exposure during fetal development. The method has a well-established prediction model, but it has not undergone formal validation and it has not been part of a ring trial. It has been used in the screening of several medium-sized compound libraries. According to the readiness criteria as published by Bal-Price et al. (2018) the neural crest cell migration assay obtained the readiness score A-.

General information

Name of test method

Circular migration inhibition of NCC (cMINC) assay, UKN2

Version number and date of deposition

This is Version 2.0 of the protocol “Assay to test impairment of migration of human neural crest cells (cMINC; UKN2) – V2.0). It was assembled and deposited in March 2023. A previous version was assembled in 2019 in the context of the EU-ToxRisk project (see publication Krebs et al., 2020)

Summary of introduced changes in comparison to previous version(s)

Changes compared to V1 refer mainly to the generation of the test system and the cell line used. Test procedures and parameters remain unchanged.

Assigned data base name

UKN2a_DART_NC_cMIGR_24h_02

Name and acronym of the test depositor

University of Konstanz (UKN), Germany

Name and email of contact person

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Reference to additional files of relevance

- An important reference is the DB-ALM Protocol n° 195. The original iPSC are in the meantime cultured feeder-free (see 3.2)
- Raw data file
- Data processing file

Description of general features of the test system source

Supply of source cells

The human induced pluripotent stem cell (hiPSC) line IMR90_clone_#4 has been bought from WiCell, Wisconsin in 2012 and a masterstock has been frozen. From the masterstock several working stocks have been prepared. The working stocks are regularly thawed and can be continously maintained due to self-renewal and pluripotency capabilities of the cells. The cells are maintained up to 10 passages before a new vial of the working stock is thawed.

Overview of cell source component(s)

Undifferentiated hiPSC cells (IMR90, WiCell) are maintained as monoculture on Laminin-521 coating in essential 8 (E8) medium. The cells grow in colonies, and are split weakly. The cells show self-renewal and pluripotency characteristics (regular testing). The cells can be differentiated into several different cell types.

Characterization and definition of source cells

- **ATCC number**: CCL-186
- **Origin**: Homo sapiens, human
- **Tissue**: lung
- **Cell type**: fibroblast

- **Gender**: female
- **Morphology**: fibroblast
- **Culture properties:** adherent
- **Disease:** no disease was diagnosed
- **Age:** 16 weeks gestation
- **Ethnicity:** Caucasian
- **Expression:** Cells express the pluripotency markers Oct4, nanog and Tra-1-60

Table 1. Short tandem repeat (STR) signature of cell line confirms 100% loci homology

	ATCC (CCL-186)	University of Konstanz; Dept. of In-Vitro-Toxikology and Biomedicin
Cell line	**IMR-90**	**IMR90**
Date		09.11.2018
D5	12	12
D5'	13	13
D13	11	11
D13'	13	13
D7	9	9
D7'	12	12
D16	10	10
D16'	13	13
vWA	16	16
vWA'	19	19
TH01	8	8
TH01'	9,3	9,3
TPOX	8	8
TPOX'	9	9
CSF1	11	11
CSF1'	13	13
Amel	X	X
Amel'	X	X

Acceptance criteria for source cell population

The cells have to be pathogen-free (regular testing for mycoplasma).
The iPSC maintenance is regularly checked for expression of pluripotency markers (Oct4, Nanog, Tra-1-60) by immunocytochemistry.

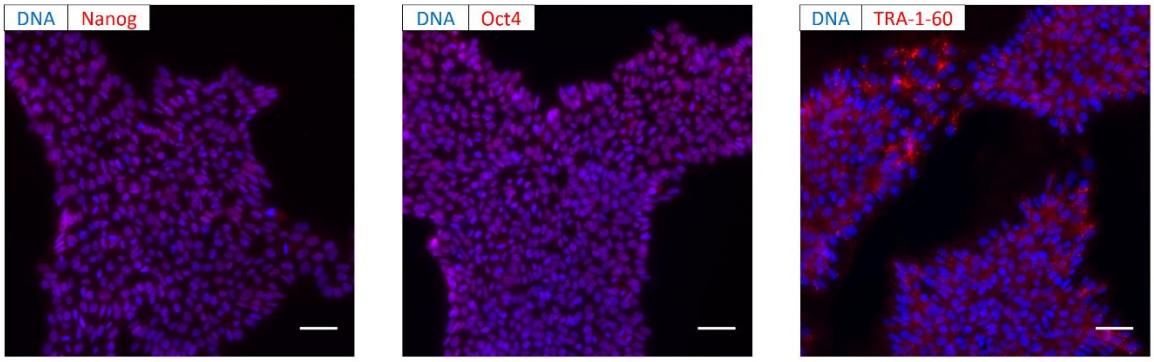


Figure 12: Immunostaining of relevant pluripotency markers in maintenance culture

Variability and troubleshooting of source cells

- hiPSC can be maintained up to 10 passages, high passage number might influence performance of cells
- Too little or too high cell density leads to detachment of cells or spontaneous differentiation
- If cells start to differentiate, cells should be discarded immediately
- Cells have to be maintained as colonies and not as single cells. Therefore splitting should be performed as fast as possible
- Batch effects of critical additives (e.g. holo-transferrin for iron supply of cells or TGF-β which maintains cell pluripotency) can lead to differentiation of cells at low passage number.

Differentiation towards the final test system

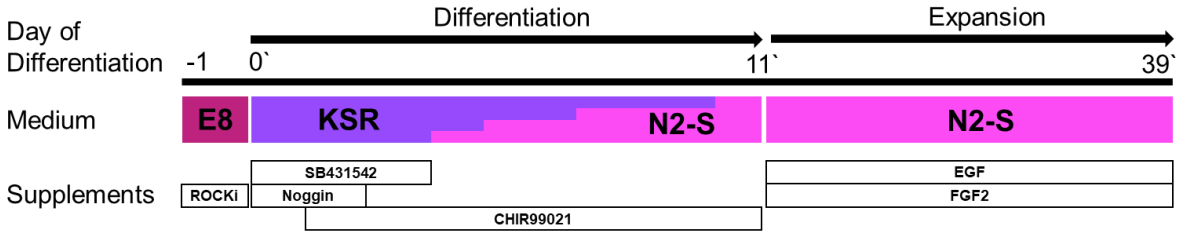


Figure 2: Differentiation scheme from hiPSCs to neural crest cells

NCCs were differentiated from hiPSCs following the modified protocol of Mica et al. (2013) (https://doi.org/10.1016/j.celrep.2013.03.025). Therefore, IMR90 iPSCs were plated on Matrigel coated 6-well plates at a density of 100'000 cells/cm^2 in E8 medium containing 10 µM ROCK (Rho-associated protein kinase)-inhibitor Y-27632. After one day cells reached a confluency of 80-100% and differentiation was initiated (day 0') by a medium change to KSR medium (Knock out DMEM (Dulbecco's Modified Eagle Medium), 15% knock out serum replacement, 1% GlutaMax, 1% MEM NEAA (minimum essential medium - non-essential amino acids) solution, 50 µM 2-mercaptoethanol) supplemented with 20 ng/ml Noggin and 10 µM SB431542. From day 2 on cells were treated with 3 µM CHIR99021. Noggin and SB431542 were withdrawn at day 3 and 4, respectively. Beginning at day 4, the KSR medium was gradually replaced with N2-S medium (DMEM/F12 (Dulbecco's Modified Eagle Medium/Ham's F-12), 1.55 mg/ml glucose, 1% GlutaMax, 0.1 mg/ml apotransferin, 25 µg/ml insulin, 20 nM progesterone, 100 µM putrescine, 30 nM selenium) in 25% increments. Cells were collected at day 11, resuspended in N2-S medium supplemented with 20 ng/ml EGF (epidermal growth factor) and 20 ng/ml FGF2 (fibroblast growth factor 2) and seeded as

droplets (10 µl) on poly-L-ornithine (PLO)/Laminin/Fibronectin coated 10 cm dishes. Cells were expanded by weekly splitting. From now on seeding as droplets was not necessary and medium was changed every second day. After 35-39 days, cells were cryopreserved at a concentration of $4*10^6$ cells/ml in 90% N2-S medium and 10% dimethyl sulfoxide (DMSO) (Merck Millipore) until further use.

Reference / link to maintenance culture protocol

DB-ALM Protocol n° 195

Definition of the test system as used in the method

Principles of the culture protocol

A highly homogeneous pre-differentiated population of neural crest cells is added to coated wells. In the middle of the well is a silicone stopper that prevents cells from settling in a circular area in the middle of the well. The cells are kept viable and alive by the presence of EGF and FGF in the medium, and they become adherent overnight. When the stopper is removed, the cells form a dense monolayer in the culture dish, with a sharply demarcated circular area in the middle that is free of cells. Due to their natural spontaneous migration behaviour, the cells move into the cell-free area. The cells are still in a proliferative state. Proliferation at the edge of the cell free area contributes to a small extent to “apparent migration behaviour”. This has been characterized and quantified in detail (Nyffeler et al. 2016). It is accounted for by counting cells in a narrower circle than the original circular area.

Acceptance criteria for assessing the test system at its start

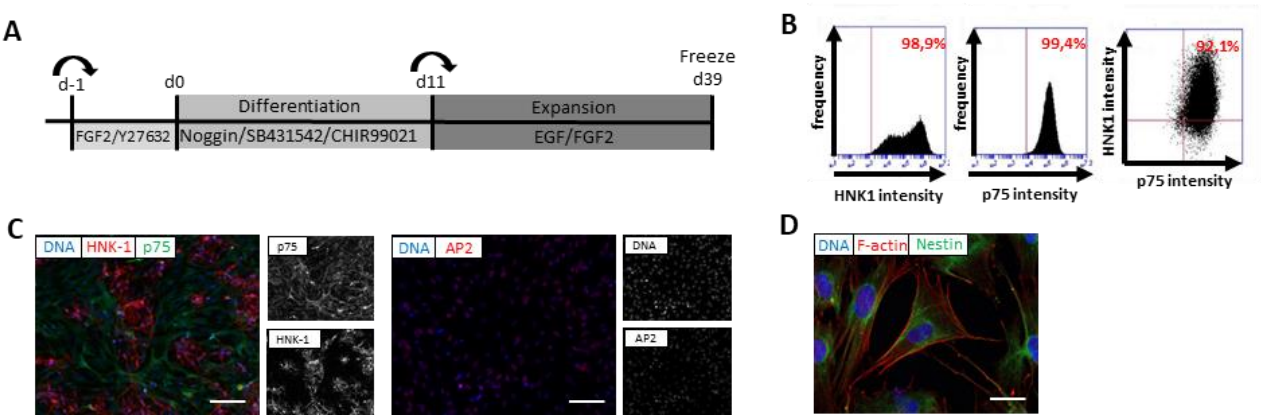


Figure 3: Characterization of the cellular system.

(A) Differentiation schema of NCCs from human induced pluripotent stem cells. (B) Expression of the NCC markers HNK-1 and p75 was monitored by FACS analysis. (C) After thawing the cells were immunofluorescently labelled for the typical NCC markers p75, HNK1 and AP2. Scale bar: 50 µm. (D) Typical mesenchymal like cell morphology of NCCs. Cells were double-stained for nestin and f-actin. Scale bar: 10 µm.

The derived NCCs are ≥ 90% positive for the NCC marker human natural killer-1 (HNK1) and the nerve low affinity nerve growth factor receptor p75. Additionally, the cells express the stem cell microfilament protein nestin and AP2. On the other hand, none of the cells show expression of the astrocyte marker GFAP, the

central nervous system precursor cell marker Pax6 and the neuronal marker βIII tubulin. Furthermore, the functional capability of the cells is checked by performing the cMINC assay with endpoint specific controls. The cells have to be pathogen-free to be used in further experiments (regular testing for mycoplasma after thawing). After thawing the viability should be > 90%.

Acceptance criteria for the test system at the end of compound exposure

After compound treatment, the negative controls should fulfil the following:

- cells should be migrated into the cell free area

- cell viability should be > 90%

Variability of the test system and troubleshooting

Causes of variability:

- High passage number of iPSC maintenance might influence NCC differentiation
- Too little cell density at the start of differentiation can cause problems (confluency should be 80-100%)
- For the freeze and thaw processes a fast handling is important to avoid cells to stay at 10% DMSO for too long. This can decrease number of viable cells after thawing.

Metabolic capacity of the test system

No specific information available.

Omics characterization of the test system

Transcriptomics data (unpublished) will become available from the originator lab (Leist) upon request.

Features of the test system that reflect the *in vivo* tissue

- The cells are able to migrate mostly as single cells; they show some evidence of collective migration
- Cells show invasive behaviour in a 3D environment (e.g. transwell assay)

Commercial and intellectual property rights aspects of cells

The cells are not protected by patents or any other licences.

Reference / link to the culture protocol

Brief description is in section 3 of this file.

Detailed maintenance is described in the DB-ALM SOP n° 195 available at:

http://cidportal.jrc.ec.europa.eu/ftp/jrc-opendata/EURL-ECVAM/datasets/DBALM/VER2-0/online/DBALM_docs/195_P_cMINC.pdf

A lab-internal handling protocol is also available upon request to the Leist-lab.

Test method exposure scheme and endpoints

Exposure scheme for toxicity testing

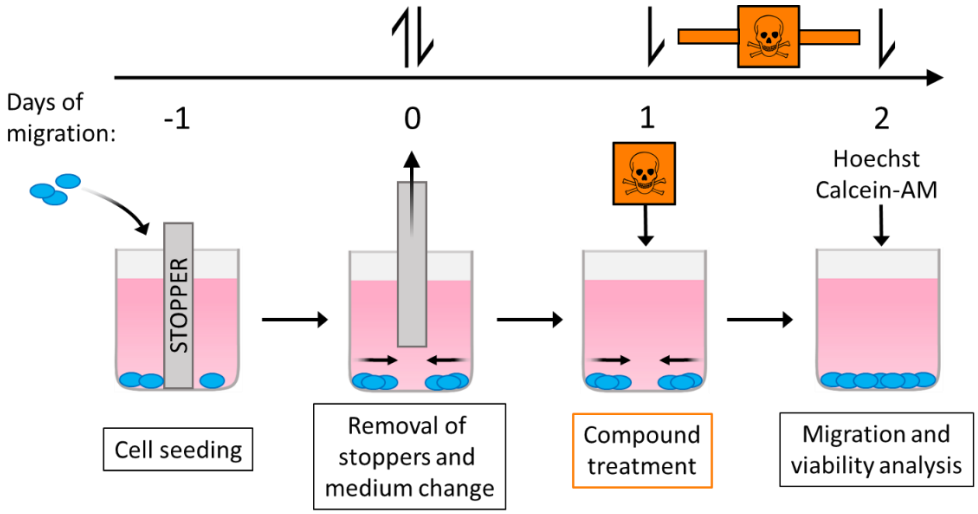


Figure 4: Exposure scheme and assay procedure.

Day -1: The wells are coated with PLO/laminin/fibronectin. Then, silicone stoppers are placed into the wells of a 96-well plate. Cells are seeded around the stoppers and allowed to attach.
Day 0: The silicone stoppers are removed and the medium is replaced with pre-warmed, fresh N2-S medium containing the cytokines EGF and FGF. Cells are allowed to migrate into the cell free area.
Day 1: 25 µl of the 5x concentrated toxicants are added to the wells. The cells plus toxicants are incubated for 24 h.
Day 2: Cells are stained with calcein-AM and Hoechst (H-33342) for 30 min before imaging with a high content imaging microscope. Quantification of migration and viability is done by high content imaging analysis.

Endpoint(s) of the test method

Specific Endpoint: Migration inhibition
Reference endpoint: Cell viability
Note: Migration induction is also possible to assess, but complex in its toxicological interpretation. Therefore only unidirectional migration is assessed as an endpoint if the method is used in a standardized testing procedure in the prediction model.

Overview of analytical method(s) to assess test endpoint(s)

Migration inhibition: NCCs are plated around silicone stoppers in a culture dish and are allowed to migrate into the cell free area upon removal of the stopper. The number of migrated cells into the cell free zone is quantified 24 h after toxicant treatment. Migration inhibition of NCCs after treatment with toxicants is measured relative to control conditions (solvent control cells). For the quantification the cells are stained with calcein-AM and H-33342 for 30 min at 37 °C. The center of the well (migration zone) is imaged in four tiles with a 5x objective. Afterwards the four images are stitched together to obtain one image. For migration quantification, a software tool has been developed (http://invitrotox.uni-konstanz.de/). With the help of this software the previously cell-free area can be estimated and the number of H-33342 and calcein double-

positive cells in the region of interest (ROI) can be counted. The diameter of the ROI was chosen so that 150 to 300 cells were in the ROI in untreated conditions. An Excel table containing the number of viable cells in the ROI of all wells of the plate is generated by the software.
Cell viability: Cell viability is measured after 48 h outside the migration zone in the same well. The cells are stained with calcein-AM and H-33342 and four fields outside the migration zone are imaged with a 10x objective. Viability is defined as the number of H-33342 and calcein double-positive cells, viable cells are determined by an automated algorithm. An excel file is generated with the number of viable cells in each well. Migration and Viability are normalized to untreated controls.

Technical details (of e.g. endpoint measurements)

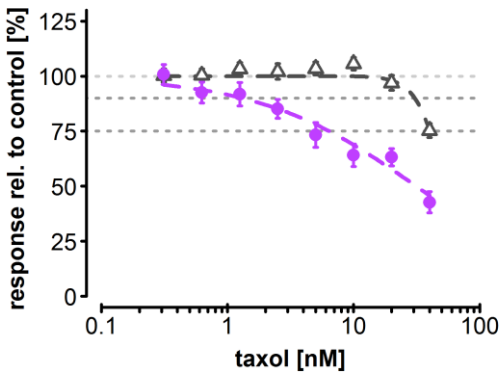
Quantification of migration
An automated microplate reading microscope (Array-ScanII HCS Reader, Cellomics, PA) equipped with a Hamamatsu ORCA-ER camera (resolution 1024 x 1024; run at 2 x 2 binning) was used for image acquisition. Four fields per well were imaged. Images were recorded in 2 channels using a 5x objective and excitation/emission wavelengths of 365 ± 50/535 ± 45 to detect H-33342 in channel 1 and 474 ± 40/535 ± 45 to detect calcein in channel 2. Pictures were exported from the microscope and for migration quantification, a software tool has been developed (http://invitrotox.uni-konstanz.de/). With the help of this software the previously cell-free area can be estimated and the number of H-33342 and calcein double-positive cells in the region of interest (ROI) can be counted. The diameter of the ROI was chosen so that 150 to 300 cells were in the ROI in untreated conditions. An Excel table containing the number of viable cells in the ROI of all wells of the plate is generated by the software.

Quantification of individual viable cells by imaging
For a quantitative assessment of viable cells, the same wells that were used to assess migration were analyzed using another image analysis algorithm of the ArrayScan VTI 700 Series software (v. 7.6.2.4) as described earlier in Stiegler et al. (2011) (doi: 10.1093/toxsci/kfr034) and Krug et al. (2013) (doi: 10.1007/s00204-013-1072-y). Nuclei were identified in channel 1 as objects according to their size, area, shape, and intensity. Nuclei of apoptotic cells with increased fluorescence were excluded. Viable nuclei were defined as H-33342 and calcein double-positive cells. An Excel table containing the number of viable cells is generated.

Endpoint-specific controls / mechanistic control compounds (MCC)

Example 1: Cell migration requires dynamic variability of the cytoskeleton, e.g. actin reorganisation. Cytochalasin D is a known inhibitor of actin polymerisation and therefore inhibits cell migration at non-cytotoxic concentrations.

Example 2: Migration depends on the dynamic instability of microtubules in the leading edge of a migrating cell. Taxol inhibits breakdown of microtubules and leads to migration inhibition at non-cytotoxic



concentrations.

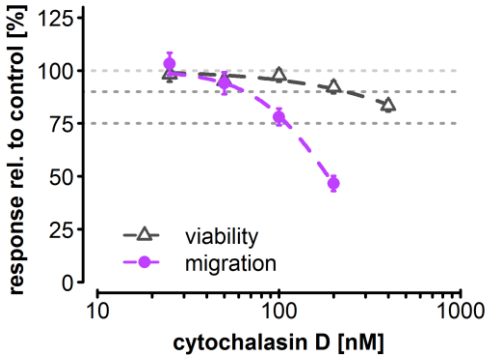


Figure 5: Examples of endpoint specific controls in the cMINC assay

Positive controls

Positive control: Cytochalasin D (200 nM), LiCl (10 mM), taxol (10 nM)

Negative and unspecific controls

Negative control: solvent (0.1% DMSO final concentration), paracetamol, ASS

Features relevant for cytotoxicity testing

Cells still proliferate after thawing and proliferation can interfere with the migration result. About 30% of the cells undergo mitosis during the assay.

Acceptance criteria for the test method

A test is discarded if the positive control did not inhibit migration, e.g. if the cell number in the migration zone is ≥ 75% of control.
If the negative control decreases migration ≥ 10%, the test is discarded.

Throughput estimate

Data point = one biological replicate (→ usually 4 technical replicates); each concentration/condition of a compound counts as a data point.
768 data points per month
2 compounds per plate, 6 different concentrations of each compound per plate (see figure 6) → 12 data points (1 plate)

16 plates can be done per week → correlates to 32 compounds → 192 data points per week
4 weeks per month → 768 data points per month.

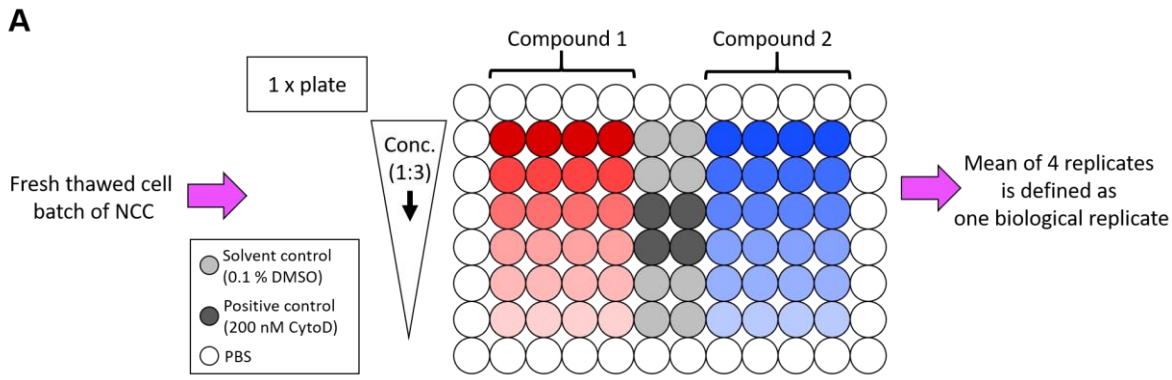


Figure 6: A typical 96-well plate layout of the cMINC

Handling details of the test method

Preparation / addition of test compounds

- Compounds are stored according to the manufacturer's instructions (e.g., 4°C, room temperature, -20°C).
- Preferable solvent is DMSO. The used DMSO is stored in a lightproof, air-tight bottle at room temperature.
- Final DMSO concentration on the cells is 0.1%
- After dissolving the compounds which are delivered in a solid/powder form, all compound solutions are aliquoted into volumes sufficient for one experiment (i.e., one biological replicate). In this way repeated freezing and thawing and therefore damaging the compound's stability and efficiency can be avoided.
- For conducting an experiment, a compound aliquot is thawed and diluted with 'DMEM/F12 Advanced' without supplements in a separate deepwell-plate.
- All compound dilutions in the master plate contain 0.5 % DMSO, so that a final concentration of 0.1 % DMSO is reached on the cells. The highest compound concentration is diluted with medium 1:200 without DMSO as 0.5 % is already reached with the DMSO the compound is solved in, the serial dilution is done with N2-S medium supplemented with 20 ng/ml EGF and FGF and 0.5 % DMSO.
- 25 µl of the 5x concentrated compound dilutions are added to the cells using a multichannel pipette.

Day-to-day documentation of test execution

Plate maps are defined prior to the experiment and documented in the lab book and files (Excel files) are stored on the work group server.
Concentrations and compound dilutions are calculated prior to the experiment.
Experimental procedures are noted manually in a paper lab book.

Practical phase of test compound exposure

The experimenter plans the experiment according to Cellomics microscope availability (has to be booked in advance).
Pipetting errors are marked directly on the plate maps and are documented in the lab book.
The paper lab book is taken to cell culture rooms and errors are documented in there right away.
The technical replicates were pipetted from left to right. The highest concentration is located at the top row.

Concentration settings

2 compounds per plate with 6 concentrations

As default a serial dilution 1:3 is used, i.e., a concentration range from e.g. 100 µM → 0.4 µM.
Serial dilutions of compounds are prepared in a separate deepwell-plate, from which 25 µl are transferred to the according plates with attached cells using a multichannel-multistepper pipette. N2-S medium supplemented with 20 ng/ml EGF and FGF is used for dilution.
Dilution steps can be adapted to be more narrow (e.g., 1:1.5)

Uncertainties and troubleshooting

- Compound solubility in stock and during dilution is too low (stock solved in 100% DMSO, final concentration of the solvent on the cells is 0.1% DMSO)
- Some compounds show autofluorescence and interfere with the detection of calcein-AM or H-33342.
- To prevent negative edge effects, only the inner 60 wells of a 96-well plate are used and the edge wells were filled with PBS.
- Focusing failure of Array Scan VTI HCS Reader (Cellomics, PA) can be a problem that produces outliers; as well as imaging only one channel.
- Highly trained/automated handling with multichannel and multistepper pipette is necessary to achieve little variance.
- Different cell batches vary in the cell number due to freezing conditions and the differing proliferation rates of the lots. The variation between the plates of one experiment is lower than 5% and the variation between different experiments is lower than 20%. The results were always normalized to untreated controls.
- Operators can get trained within 2-4 weeks. Cell seeding and medium change should be performed as fast as possible to keep cells as short as possible at room temperature. The more practice an operator has, the faster the critical steps can be performed.
- Substances are added when pipette tips are touching the wall of the wells right above the medium surface. When the substance solution is pipetted too high above the medium surface, the droplet may just stick to the wall of the well without flowing down into the medium.

Detailed protocol (SOP)

Protocol n° 195 in DB-ALM data base

http://cidportal.jrc.ec.europa.eu/ftp/jrc-opendata/EURL-ECVAM/datasets/DBALM/VER2-0/online/DBALM_docs/195_P_cMINC.pdf

Updated SOP can be made available by laboratory upon request:
marcel.leist@uni-konstanz.de

Special instrumentation

- The method requires a Cellomics Array Scan VTI HCS high content reader that may not be present in the standard lab. (Alternative automated microscopes and software tools to perform migration assays exist and the method might be transferred in the future)
- Silicone stoppers (Platypus Technologies, Madison, WI, US).

Possible variations

a) further additional endpoints:
- EdU (5-Ethynyl-2′-deoxyuridine) staining can be performed within the assay to measure proliferation

b) other analytical endpoints:
cell viability by:
- fluorimetric measurement of resazurin conversion
- measurement of extracellular LDH *(lactate dehydrogenase)*
- measurement of luminescence indicating ATP (*adenosine triphosphate)* content

c) other exposure:
- The exposure time for toxicants can be increased up to 48 h. But longer exposure time increases the effect of toxicants on viability and cell proliferation.

d) experimental variation:
- AraC (cytosine arabinoside) addition to prevent proliferation effects on migration result. About 30% of the cells divide during the 24 h of the assay period. Therefore a cell proliferation inhibiting compound would reduce the cell number by 25% in the migration zone and therefore results in 80% viability and 75% migration measurement. But a reduction of migration by > 25% is unlikely to be explained by effects on proliferation (Nyffeler J. et al., 2016).
- other cells may be used (derived from different iPSC)

Cross-reference to related test methods

The scratch assay is another method to analyse cell migration. In comparison to the cMINC assay the throughput of the scratch assay is low (Zimmer B. et al., 2012).

Data management

Raw data format

Raw data is extracted by copy-paste in Excel files (example file available upon request).
Data from all technical replicates are collected in one file.

Outliers

1. Mathematical procedures to define outliers have not been defined. Data points that are far off (i.e. more than the known endpoint variability which would be 25% for migration) are discarded. Biological outliers do practically not exist, most far data points are the result of technical problems (focus not found, only one channel imaged, etc.)
2. All raw data (incl. outliers) are stored.
3. Technical outliers make up 1-0.1%.

Raw data processing to summary data

- Array Scan VTI HCS Reader (Cellomics, PA) takes images (optionally bitmap or tiff-format; 512 x 512 pixels, 8bit or 16bit)
- For migration quantification images from the Array Scan VTI HCS Reader are exported in 8-bit tiff format and loaded to the “Ringassay” software (http://invitrotox.uni-konstanz.de/) for calculating the number of migrated cells.
- Images are locally analyzed using the Array Scan software, algorithm quantifies cell count (viable cells)
- data are copy-pasted into an Excel sheet, further analysis is done with Excel + GraphPad Prism

Curve fitting

The data are analyzed with Excel and represented with GraphPad Prism.
For the concentration curve, a nonlinear regression fit is calculated. The fitting method is least squares. If a non-linear curve fit is not possible, a linear curve fit is performed. The curve deriving from the fit is a 4-parameter log function. To calculate the EC25 value, this log-function is solved for y=25% of the total scale, not for 25% of the min-max scale (see example below). Treated concentrations are analyzed for deviation from control. Sometimes it is analyzed whether the deviation of neurite growth is different from the deviation of viability via comparison of the means ± deviation using two-way ANOVA + Tukey-Kramer post hoc testing. Significance levels of data points compared to solvent control are determined via one-way ANOVA followed with non-parametric Dunnett's post test.

BMC values with their upper and lower confidence intervals (BMCU and BMCL) are calculated via the publically available online software:
http://invitrotox.uni-konstanz.de/BMCeasy/

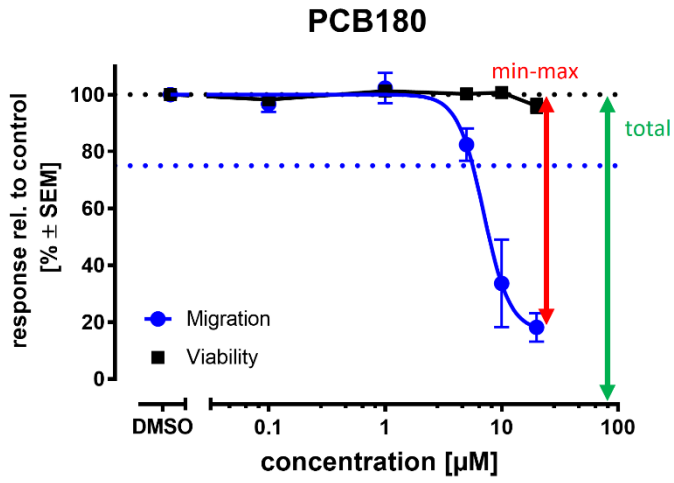


Figure 7: Exemplification of curve fit and relative effect responses

Internal data storage

The data are firstly stored on the microscope computer and then exported to other servers (lab group server and university server), which are back-upped regularly.

Metadata

The metadata are documented, stored and exported as text document (log)-files to the according scheme: (local PC)_descriptor(date and time)_XXX.log:
The following metadata are stored:

- cellinsight-pc_160429130003_AutomationControllerIni
- cellinsight-pc_160429130003_kineticprotocol
- cellinsight-pc_160429130003_protocol
- cellinsight-pc_160429130003_scan
- cellinsight-pc_160429130003_ScanIni
- cellinsight-pc_160429130003_spooling
- cellinsight-pc_160429130003.spooled

Metadata file format

Metadata files are available.

Prediction model and toxicological application

Scientific principle, test purpose and relevance

Migration of NCCs is an essential process during fetal development. Impaired NCC migration triggered genetically or by toxicants can lead to malformations and disorders e.g. Hirschsprung's disease or Treacher-Collins syndrome.
The cMINC assay models the effects of NCC migration under toxicant exposure.
The test captures endpoints like spina bifida or cleft palate also measured during developmental toxicity regulatory studies.

Prediction model

Three different models are used:

1. prediction model for screening:
hit = inhibition of NCC migration while viability is not changed:
Migration ≤ 80% of DMSO control
Viability ≥ 90% of DMSO control

2. prediction model for compound hazard evalaution:
hit confirmation testing: EC10 Viability (V) / EC 25 Migration (M) ≥ 1.3 → specific migration inhibitor of NCCs

3. prediction model for borderline compounds:
A ratio of BMC10 Viability (V) / BMC**L**25 Migration (M) ≥ 1.3 is considered a borderline hit. In some scenarios the viability does not reach the BMC10 Viability (V) necessary for the ratio calculations. In this case the highest tested concentration (HTC) was used. Schematic representation of the complete prediction model is shown in Fig. 8 below.

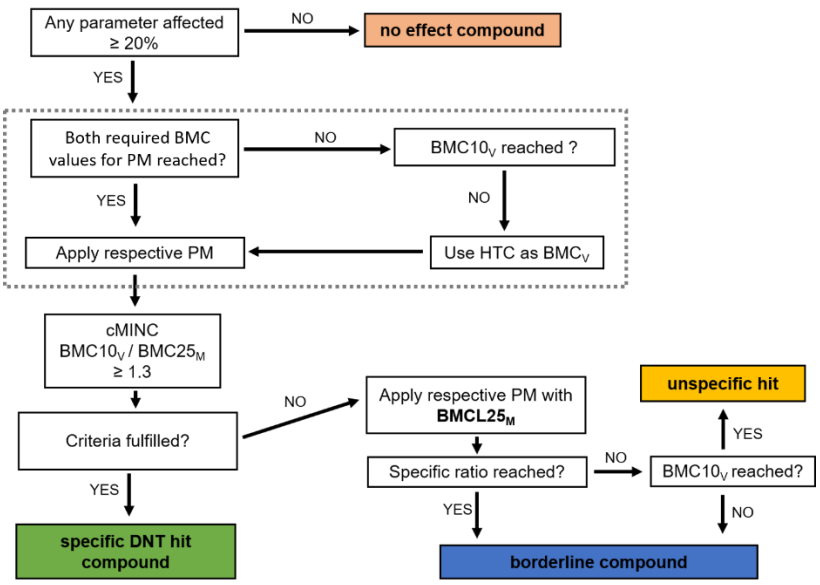


Figure 8: UKN2 Prediction model classification tree

A full overview with a schematic representation of the UKN4 prediction model can also be found in supplementary figure 3 of Blum et al., 2022.

Prediction model setup

a) The prediction model was established using the following compounds (Nyffeler et al. 2016):
- acrylamide
- arsenic trioxide (As_2O_3)
- cadmium chloride ($CdCl_2$)
- lithium chloride (LiCl)
- PCB180
- retinoic acid

➔ Positive controls

- cytochalsin D
- taxol
- colchicine

➔ Endpoint-specific controls

- silver nitrate ($AgNO_3$)
- cytarabine (AraC)
- aphidicolin
- L-homocysteine
- MG-132
- staurosporine
- triton X-100

➔ Unspecific toxicants

b) The prediction model has been applied to screen the 80 compound library of NTP (Nyffeler et al., 2017). The prediction model including the borderline classification has been applied to screen a 120 compound library (Blum et al., 2022; Masjosthusmann et al., 2020).

c) The process is documented in Nyffeler et al. 2016

d) Sensitivity and specificity require still definition of a good standard and have therefore not been done.

Test performance

Some background on the test performance is given in chapters 8.2/8.3 (prediction model).

Several performance parameters for the test were obtained in several separate evaluation rounds.

A first evaluation was done during the first publication of the model and its applications (Nyffeler et al. 2016). Here, a panel of well-selected positive and negative controls have been tested. Accordingly, the specificity was 100% and the sensitivity was > 90 %. In dedicated experiments, S/N ratios of > 20 and a z' of > 0.5 have been determined. The compound used as positive control cytochalasin D was run across 35 different assay plates. The migration percentage relative to the solvent control varied between 30 and 70% across all plates (Nyffeler et al. 2016, Supplementary S2).

Later, the test has been used in screening campaigns, and real-live performance data under broader screen conditions have been obtained. The different performance data need to be considered, when a compound is a hit in a screen, or whether it has been specifically evaluated in a hit follow-up or a mechanistic project.

A first screen application has been the NTP80 screen (80 compounds provided by the US NTP). Data are published in Nyffeler et al. 2017.

A second screen application has been the cross systems case study of the EU-ToxRisk project. The baseline variation is indicated in Krebs et al., 2020. Moreover, an overview is given for 19 compounds on the BMC/BMCL ratio as measure of readout certainty.

A third screen was performed in the context of the EFSA DNT test battery evaluation with 120 compounds (Masjosthusmann et al. 2020). From this screen the following performance indicators were obtained:

A: **Specificity of DNT IVB: 100%**

→ Also with cMINC as standalone assay in 17 `tool negatives´ tested (Masjosthusmann et al. 2020).

B: **Sensitivity of DNT IVB: 82.7%**

→ With cMINC combined in a full DNT battery and a selected set of 27 positive compounds with evidence for DNT (Masjosthusmann et al. 2020).

C: **Baseline variation (intra-experimental)**

Migration: 14.8 ± 4.3%

Cell viability: 6 ± 3%

D: **Baseline variation (inter-experimental)**

Migration: 15.6%

Cell viability: 9.7%

E: **Variation of a positive control run on each (inter-experimental)**

Migration: 32.2%

Definition of values C-E

C: **Baseline variation (intra-experimental)** is the mean coefficient of variation CV±SD of the CV of all replicates of the solvent control from each experiment across n>200 experiments.

D: **Baseline variation (inter-experimental)** is the variability across all independent experiments (n>200) after normalization based on the response of the lowest test concentration. It was assumed that the lowest test concentration does not affect any of the endpoints measured.

E: **Variation of a positive control run on each (inter-experimental)** is the variability of the positive control

across all independent experiments (n>40) after normalization. Example for a positive control that on average reduced the specific endpoint down to 40% (relative to solvent control) and a calculated variability of 50%: 0.5 x 40% = +/- 20% → The positive control with mean of 40% varies from 20% to 60%.

In vitro – in vivo extrapolation (IVIVE)

1. Estimated lipid content and albumin concentration in in vitro test media and human plasma:

Medium	Lipid content (ml/ml)	Albumin concentration (µM)
UKN2	2.8E-6	5.6
Human plasma	6000	600

2. A three-step (physiology-based) pharmacokinetic (PBPK) modelling strategy has been used to evaluate the clinical relevance of the in vitro concentrations, which impair NCC migration in the scratch assay. The in vivo plasma concentrations of the tested compounds were within the same range as the concentrations used in the scratch assay (Zimmer B. et al., 2014).

3. No special considerations known.

Applicability of test method

Pesticides, flame retardants, polychlorinated biphenyls (PCBs), drugs have been detected.
The exact applicability domain is not yet clear.
Volatile compounds and substances that are not water-soluble cannot be measured.

Incorporation in test battery

a) Strengths:

- Medium throughput (The term "medium" is derived from a comparison and in relation to a number of other known DNT NAMs)
- Automated microscopy

b) The comparable scratch assay is a low throughput assay. The orthogonal transwell-assay is also of lower throughput.
c) The assay has been considered as part of the ESNATS (Embryonic Stem cell-based Novel Alternative Testing Strategies) screen battery.
d) The test can to some extent be stand-alone (positive hits are meaningful). The 'negative hits' provide little information and require other tests in a developmental toxicity test battery

Publication / validation status

Availability of key publications

Establishment of a human cell-based in vitro battery to assess developmental neurotoxicity hazard of chemicals
Blum, J. et al. Chemosphere, 2022. PMID: 36328314

Neurodevelopmental toxicity assessment of flame retardants using a human DNT in vitro testing battery.
Klose, J. et al. Cell Biol Toxicol. 2021. PMID: 33969458

Establishment of an a priori protocol for the implementation and interpretation of an in-vitro testing battery for

the assessment of developmental neurotoxicity.
Masjosthusmann, S. et al. EFSA Supporting Publications. 2020; 17(10): 1938E.

The EU-ToxRisk method documentation, data processing and chemical testing pipeline for the regulatory use of new approach methods.
Krebs, A. et al. Arch. Toxicol., 2020. PMID: **32632539**

A structure-activity relationship linking non-planar PCBs to functional deficits of neural crest cells: new roles for connexins.
Nyffeler, J. et al. Arch. Toxicol., 2018. PMID: 29164306

Combination of multiple neural crest migration assays to identify environmental toxicants from a proof-of-concept chemical library.
Nyffeler, J. et al. Arch. Toxicol., 2017. PMID: 28477266

Design of a high-throughput human neural crest cell migration assay to indicate potential developmental toxicants.
Nyffeler, J. et al. ALTEX, 2016. PMID: 27463612

Identification of transcriptome signatures and biomarkers specific for potential developmental toxicants inhibiting human neural crest cell migration.
Pallocca, G. et al. Arch.Toxicol., 2016. PMID: 26705709

Profiling of drugs and environmental chemicals for functional impairment of neural crest migration in a novel stem cell-based test battery.
Zimmer, B. et al. Arch. Toxicol., 2014. PMID: 24691702

Evaluation of developmental toxicants and signaling pathways in a functional test based on the migration of human neural crest cells.
Zimmer, B. et al. Environ. Health Perspect., 2012. PMID: 22571897

(Potential) linkage to AOPs

Test method is not linked to AOP.

Steps towards mechanistic validation

a) Cells express typical NCC markers, are of human origin and are able to migrate
b) Cell migration requires dynamic variability of the cytoskeleton, e.g. actin reorganization and the dynamic instability of microtubules in the leading edge. If this is inhibited by cytochalasin D or taxol migration of NCCs is inhibited.
c) A formal mechanistic validation has not been performed.

Pre-validation or validation

To date, 141 unique compounds (as defined by unique DTXSIDs) have been tested successfully in this assay. No formal OECD 34 validation study has been done (e.g., ring trials with a standard set of known positive and negative controls).
In total, >200 different compounds were tested in the cMINC assay. The test method was developed using a compound training set (Nyffeler et al. 2016). It was used for an 80 compound screening library from the US National Toxicology Program (NTP) (Nyffeler et al. 2017). The test method was part of a DNT hazard assessment for 120 compounds in a DNT testing battery. The later compound set includes potential DNT

positive and DNT negative compounds (Blum et al., 2022).

Linkage to (e.g. OECD) guidelines / regulatory use

Test is not linked to regulatory guidelines.

Test method transferability

Operator training

Experiences are required in:
- cell culture
- multichannel/multistep pipetting
- handling of Array Scan VTI HCS Reader (Cellomics, PA) and its software
- Microsoft Excel
- Ringassay software
- GraphPad Prism

Operator is trained and guided by a highly experienced instructor. Approximately 4 weeks will be needed for a smooth assay performance.
Learning iPSC culture and cell differentiation takes several months.

Transfer

The assay hasn't been transferred or applied in other labs.

Safety, ethics and specific requirements

Specific hazards; issues of waste disposal

No specific requirements.

Safety data sheet (SDS)

SDS are available in the university of Konstanz DaMaRIS database (**Da**ngerous **Ma**terials **R**egistry **I**nformation **S**ystem).

Specific facilities / licenses

Work requires S1 cell culture laboratories (genetically modified cells).
No specific facilities are required.
No specific ethical approval is required.

Commercial aspects / intellectual property of material / procedures

To our best knowledge, no elements needed to conduct the experimental part of the test method are protected. Programs used to conduct the analysis of the data (Microsoft Excel and GraphPad Prism) need to

be purchased or obtained by license agreement, however data analysis and plotting can be done with other, freely available tools.